

Diebndu Singh

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Professional Summary

Ph.D. candidate in Chemistry specializing in structural biology and molecular motor proteins, with a focus on DNA translocase SpoIIIE. Experienced in cryo-EM data processing (CryoSPARC), protein purification, and enzymatic kinetics (ATPase, EMSA). Investigating the mechanistic relationship between oligomerization, structure, and ATPase activity in DNA translocation systems.

Education

PhD in Chemistry <i>Indiana University Bloomington</i>	<i>July 2024 – June 2029</i>
MS in Chemistry <i>Indian Institute of Technology Bombay</i>	<i>July 2022 – June 2024</i>
BS in Chemistry <i>University of Calcutta</i>	<i>July 2019 – June 2022</i>

Technical Skills

Structural Biology: Cryo-EM data processing and single-particle analysis (CryoSPARC), model building (ChimeraX, PyMOL)

Biochemistry: Recombinant DNA Technology, Gene Cloning, Site-Directed Mutagenesis, Protein Expression (E. coli, BL21 Star).

Purification: FPLC (AKTA), Affinity Chromatography (Ni-NTA), Size Exclusion (SEC), Ion Exchange.

Software/Tools: CryoSPARC, ChimeraX, PyMOL, Python, Shell Scripting (Linux/macOS), LaTeX.

Research Projects

Graduate Student <i>Marc C. Morais Lab, Indiana University Bloomington</i>	<i>Indiana, IN</i> <i>August 2024 – Aug 2029</i>
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- Investigating the molecular mechanism of DNA translocation in SpoIIIE using integrated structural and biochemical approaches
- Purified recombinant SpoIIIE and identified higher-order oligomer (650 kDa) using SEC, indicating stable multimer formation
- Measured ATPase activity across varying protein concentrations and observed loss of cooperative behavior at lower concentration, suggesting oligomerization-dependent catalysis
- Processed cryo-EM datasets in CryoSPARC (motion correction, 2D classification, 3D reconstruction), identifying filamentous assemblies distinct from expected ring structures
- Optimized purification and assay conditions to improve protein stability and reproducibility of higher-order assemblies

MS Project <i>Prof. Ishita Sengupta, IIT Bombay</i>	<i>Mumbai, India</i> <i>July 2023 – June 2024</i>
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- Developed NanoLuc-based luminescent biosensors and quantified detection sensitivity for OPS using luminescence assays
- Performed molecular cloning (homologous recombination), DNA prep/analysis, and protein purification.
- Characterized protein folding (CD) and evaluated biosensor performance using luminescence assays

Summer Internship <i>Prof. Alok Das, IISER Pune</i>	<i>Mumbai, India</i> <i>May 2023 – July 2023</i>
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- Studied intramolecular interactions ($N-H \cdots N(C5)$, $N-H \cdots S(C6)$) using gas-phase laser spectroscopy and solution-phase techniques
- Quantum Mechanical Calculation and optimization of molecule energy for tri-peptide molecule with GAUSSIAN
- Characterized protein folding (CD) and evaluated biosensor performance using luminescence/fluorescence assays

Publications

, *John Doe*,



Jan 2004